

STAYKARO AI CALLER AGENT

Team Execution Plan V1.1 — Compact & Readable Edition

Who does what · Which PDF to read · Dependencies · Reviews · Checkpoints

Shivashree · Nishkala · Nihal

Main working document. The six technical PDFs remain reference manuals.

1. How to use this plan

Start here every day. Find your current ticket, check its dependency and reviewer, then open the technical PDF listed in the ticket. The technical details stay in the six reference PDFs instead of being duplicated here.

Completion rule: code generated by a person or AI is not automatically done. Tests, review, integration evidence and acceptance criteria must pass.

2. PDF Access & Reading Matrix

Everyone may access all six PDFs. **Required** means actively read/use it. **Primary** means it directly covers your owned domain. **Reference** means open it when your ticket or dependency requires it.

PDF	Shivashree	Nishkala	Nihal	Why / use
Developer Handbook	Required	Required	Required	Git, repository, PR and coding workflow
API Specification	Required	Required / Primary	Required	API and event contracts
Database Design & Data Model	Reference	Required / Primary	Reference	PostgreSQL, Redis, pgvector, RLS
Sprint Planning Guide	Required	Required	Required	Phases, dependencies, weekly targets
Infrastructure & Operations Guide	Reference	Reference	Required / Primary	Docker, VPS, Nginx, CI/CD, monitoring
Testing Guide	Required	Required	Required	Unit, integration, voice, security and load tests

Recommended reading order

Person	Read first	Keep as reference
Shivashree	Execution Plan → Developer Handbook → API Specification → Testing Guide	Database, Sprint, Infrastructure
Nishkala	Execution Plan → Developer Handbook → Database Design → API Specification → Testing Guide	Sprint, Infrastructure
Nihal	Execution Plan → Developer Handbook → Infrastructure Guide → API Specification → Testing Guide	Database, Sprint

Phase 0: PDFs to open right now

Person	Open now	Immediate purpose
Shivashree	Execution Plan + Developer Handbook + API Specification	SH-03, provider contracts, call-session integration
Nishkala	Execution Plan + Developer Handbook + Database Design + API Specification	NK-01 → NK-04 foundation
Nihal	Execution Plan + Developer Handbook + Infrastructure Guide	NH-01 → NH-05; begin NH-16 inquiry

3. Team ownership

Member	Role	Main ownership
Shivashree	Voice & AI	Telephony, Pipecat, STT, LLM, TTS, turns, multilingual voice, latency
Nishkala	Backend, Data & RAG	FastAPI, PostgreSQL, Redis, auth/RBAC, RLS, RAG, persistence, usage/billing
Nihal	Frontend, Automation & DevOps	Dashboards, Docker/VPS/Nginx, CI/CD, monitoring, integrations

4. Phase Responsibility Matrix

Phase	Shivashree	Nishkala	Nihal
0 Foundation	S	P	S
1 Voice Pipeline	P	S	R
2 Voice Behaviour	P	S	S
3 RAG	S	P	R
4 Multi-tenancy/Security	R	P	S
5 Multilingual	P	S	R
6 Admin Dashboard	R	S	P
7 Client Dashboard	R	S	P
8 Integrations	R	S	P
9 Usage/Billing	R	P	S
10 Hardening	P*	P*	P*
11 Pilot	P*	P*	P*
12 Outbound (gated)	P	S	R

P = Primary · S = Supporting · R = Reviewer · * shared primary by domain.

5. Current Phase 0 — Engineering Foundation

Goal: a clean checkout starts the agreed stack; FastAPI reaches PostgreSQL/Redis; health/readiness passes; the voice gateway can create and clean up an empty call session.

Nishkala (Primary): NK-01 → NK-02 → NK-03 → NK-04.

Nihal (Support): NH-01 → NH-02 → NH-03 → NH-04 → NH-05. Start NH-16 licensing inquiry early.

Shivashree (Support): learn Pipecat/provider contracts and begin SH-03 once NK-01/NK-03 contracts exist.

Checkpoint 1: Docker stack starts, /health and /ready pass, Redis lifecycle passes, and empty call connects/disconnects cleanly.

6. Shivashree — Voice & AI Tickets

ID / Task	Phase / Pri	Reviewer	Depends	Done when	Read
SH-01 Exotel adapter	P1 / P0	Nishkala	SH-03	Telephony sandbox call works	API + Testing
SH-02 Twilio adapter	P1 / P1	Nishkala	SH-03	Same provider contract	API + Testing
SH-03 Pipecat pipeline + session manager	P1 / P0	Nishkala	NK-01, NK-03	Empty call lifecycle clean	Developer + API
SH-04 Deepgram streaming STT	P1 / P0	Nishkala	SH-03	Interim/final transcript streams	Testing
SH-05 Endpoint detection tuning	P2 / P0	Nishkala	SH-04	Natural measured endpointing	Testing
SH-06 Groq streaming LLM	P1 / P0	Nishkala	SH-03	Tokens stream before completion	API + Testing
SH-07 OpenAI fallback	P2 / P0	Nishkala	SH-06	Context preserved on fallback	Testing
SH-08 Streaming TTS	P2 / P0	Nishkala	SH-03, SH-06	First audio streams early	Testing
SH-09 TTS fallback chain	P2 / P0	Nishkala	SH-08	No unexplained dead air	Testing
SH-10 Barge-in	P2 / P0	Nishkala	SH-05, SH-08	Caller interruption stops AI	Testing
SH-11 Turn management	P2 / P0	Nishkala	SH-03, NK-03	turn_id + language_code events	API + DB
SH-12 Turn-level language detection	P5 / P0	Nishkala	SH-04, SH-11	Mid-call switching works	Testing
SH-13 Latency instrumentation	P2 / P0	Nihal	SH-04/06/08	Per-stage latency recorded	Testing + Infra
SH-14 Voice → RAG integration	P3 / P0	Nishkala	NK-10, SH-11	Grounded/no-match correct	API + Testing
SH-15 Multilingual voice matrix	P5 / P0	Nishkala + Nihal	SH-12, NK-11	Language matrix passes	Testing
SH-16 Failure recovery / completed event	P2 / P0	Nishkala	SH-07/09/11	Predictable fallback	Testing + API
SH-17 Concurrent voice load support	P10 / P0	Nishkala + Nihal	NH-18, SH-13	Measured load profile	Testing + Infra

7. Nishkala — Backend, Data & RAG Tickets

ID / Task	Phase / Pri	Reviewer	Depends	Done when	Read
NK-01 FastAPI skeleton/config	P0 / P0	Nihal	—	Clean API boot	Developer + API
NK-02 PostgreSQL schema/migrations	P0 / P0	Nihal	NK-01	Fresh DB migrates	Database
NK-03 Redis call sessions	P0 / P0	Shivashree	NK-01	Lifecycle + TTL cleanup	Database + API
NK-04 Health/readiness	P0 / P0	Nihal	NK-01/02/03	Dependency health accurate	API + Infra
NK-05 JWT authentication	P4 / P0	Nihal	NK-01/02	Invalid tokens rejected	API + Testing
NK-06 RBAC	P4 / P0	Nihal	NK-05	Role matrix passes	API + Testing
NK-07 Tenant context + PostgreSQL RLS	P4 / P0	Independent	NK-05/06/02	Zero cross-tenant context	Database + Testing
NK-08 Tenant isolation penetration suite	P4 / P0	Independent	NK-07	ZERO unauthorized success	Testing
NK-09 Knowledge ingestion	P3 / P0	Nihal	NK-02	Docs ready with tenant tags	Database + API
NK-10 pgvector retrieval	P3 / P0	Shivashree	NK-09, NK-07	Tenant-safe top-K/no-match	Database + Testing
NK-11 Multilingual embeddings evaluation	P5 / P0	Shivashree + Nihal	NK-10	Evidence-based model choice	Testing
NK-12 Call persistence	P2 / P0	Shivashree	NK-02, SH-11	Ordered complete transcript	Database + API
NK-13 Usage metering	P9 / P0	Nihal	NK-12	Usage reconciles	Database + API
NK-14 Razorpay billing	P9 / P0	Nihal	NK-13	Sandbox lifecycle works	API
NK-15 Webhook verification/idempotency	P9 / P0	Nihal	NK-14	Duplicate webhook harmless	API + Testing
NK-16 Audit logging	P10 / P0	Nihal	NK-05, NK-07	Sensitive changes audited	Database + Testing
NK-17 Backup/restore + hardening	P10 / P0	Nihal	NK-02, NK-07	Actual restore passes	Infra + Testing

8. Nihal — Frontend, Automation & DevOps Tickets

ID / Task	Phase / Pri	Reviewer	Depends	Done when	Read
NH-01 Docker Compose skeleton	P0 / P0	Nishkala	NK-01/NK-03/SH-03	Clean stack starts	Infrastructure
NH-02 Service Dockerfiles	P0 / P0	Nishkala	NH-01	Reproducible images	Infrastructure
NH-03 Nginx reverse proxy	P0 / P0	Shivashree	NH-01	HTTPS/WSS works	Infrastructure
NH-04 GitHub Actions CI/CD	P0 / P0	Nishkala	NH-01	Failing PR blocked	Developer + Infra
NH-05 VPS deployment scripts	P0 / P0	Shivashree	NH-01, NH-04	Repeatable staging deploy	Infrastructure
NH-06 Admin client management	P6 / P0	Nishkala	NK-05/06	Ops manages clients	API
NH-07 Admin live calls/health	P6 / P0	Shivashree	SH-13, NK-04	Latency/health visible	API + Infra
NH-08 Admin usage/cost	P6 / P1	Nishkala	NK-13/14	Values match backend	API
NH-09 Client agent config	P7 / P0	Nishkala	NK-05/06	Tenant-safe config	API
NH-10 Client knowledge base	P7 / P0	Nishkala	NK-09	Upload/status works	API
NH-11 Calls/transcripts/analytics	P7 / P0	Nishkala	NK-12	Only own tenant visible	API + Testing
NH-12 Client billing UI	P7 / P1	Nishkala	NK-14	Billing matches backend	API
NH-13 Integration service	P8 / P0	Nishkala	SH-16	Async event path works	API + Infra
NH-14 Email/WhatsApp automation	P8 / P1	Nishkala	NH-13	Sandbox follow-up works	Testing
NH-15 CRM adapter framework	P8 / P1	Nishkala	NH-13	Test CRM receives record	API
NH-16 n8n license/isolation decision	P8 / P0	Nishkala	—	Written go/no-go decision	Infrastructure
NH-17 Deployment draining	P10 / P0	Shivashree	NH-05, SH-03	Active call survives deploy	Infra + Testing
NH-18 Monitoring/alerts/load harness	P10 / P0	Shivashree + Nishkala	NK-04, SH-13	Alerts + measured capacity	Infra + Testing

9. Nine Integration Checkpoints

CP	Gate	Required result	Blocking failure
CP1	Foundation	Stack + health + empty call lifecycle	Core service unreliable
CP2	Voice + persistence	Real call → ordered transcript	Lost/misordered turns
CP3	Voice + RAG	Grounded Q&A; + safe no-match	Fabricated answer
CP4	Multi-tenancy	All cross-tenant attempts fail	ANY unauthorized success = HARD STOP
CP5	Multilingual	Voice + retrieval matrices pass	Approved language fails
CP6	Dashboards	Admin/client workflows tenant-safe	Isolation/workflow failure
CP7	Automation + billing	Async integration + billing reconcile	Live-call coupling/duplicate billing
CP8	Hardening	Load + restore + drain + alerts proven	Unverified production claim
CP9	Pilot	All P0 launch gates green	Open P0/security blocker

10. 13-week roadmap

Week	Focus	Primary work
1	Foundation	NK-01→04; NH-01→05; NH-16 inquiry
2	Voice skeleton	SH-03 + CP1
3	Real speech	Telephony + STT + LLM
4	Turns/persistence	TTS + turn events + NK-12 + CP2
5	RAG	NK-09/10 + SH-14 + CP3
6	Auth/RBAC	NK-05/06
7	Tenant isolation	NK-07/08 + CP4 HARD GATE
8	Multilingual	SH-12/15 + NK-11 + CP5
9	Admin dashboard	NH-06/07
10	Client dashboard	NH-08→12 + CP6
11	Automation/billing	NK-13→15 + NH-13→16 + CP7
12	Hardening	SH-17 + NK-16/17 + NH-17/18 + CP8
13	Pilot readiness	All three + CP9

11. Review, handoff and Definition of Done

PR flow: feature branch → tests → PR → independent review → develop → checkpoint → main. No self-approval for production-critical work.

Handoff: contract/payload example + PR + test evidence + environment variable names (never values) + known limitations + consumer confirmation.

Done: implementation + tests + error handling + logs/metrics where relevant + security/tenant check + documentation + independent review + integration/staging evidence + acceptance result.

12. Which technical PDF to open

When you need...	Open...
Git/repo/coding/PR rules	Developer Handbook
Endpoint or event contract	API Specification
PostgreSQL, Redis, pgvector, RLS, migrations	Database Design & Data Model
Detailed weekly sequencing	Sprint Planning Guide
Docker, VPS, Nginx, CI/CD, monitoring, backup/deploy	Infrastructure & Operations Guide
Unit/integration/voice/RAG/security/load tests	Testing Guide

13. Production gates and open decisions

Before pilot: zero cross-tenant successes; measured load capacity; actual backup restore; deployment drain test during active call; alerts proven; multilingual matrices green; billing webhook idempotency proven; no secrets in Git; all P0 tickets closed.

Still gated/open: n8n licensing + tenant isolation; Tamil V1 interpretation; final telephony/provider approval; caller-heard fallback UX; outbound compliance/company approval.

Current instruction: finish Phase 0 and CP1 before treating later phases as active.